Harder Geometrical Problems — problem 12.3.2.62

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(a) Every diagonal of a convex quadrilateral $ABCD$ divides its area into two equal parts. Prove that $ABCD$ is a parallelogram.

(b) The diagonals AD, BE, CF divide the convex hexagon $ABCDEF$ into two equal parts. Prove that these diagonals pass through one point.